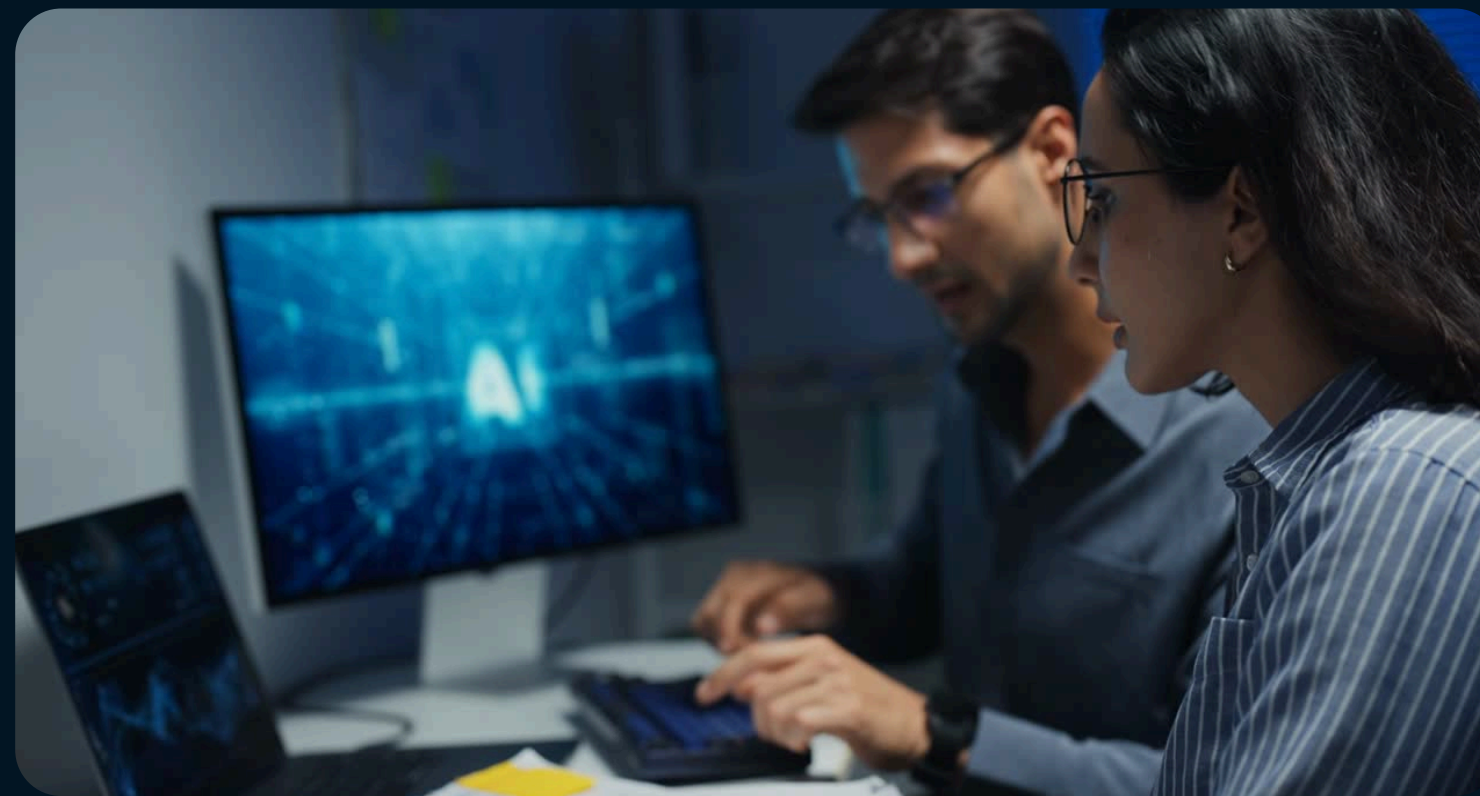


Do AI tools affect wages differently depending on developer experience?

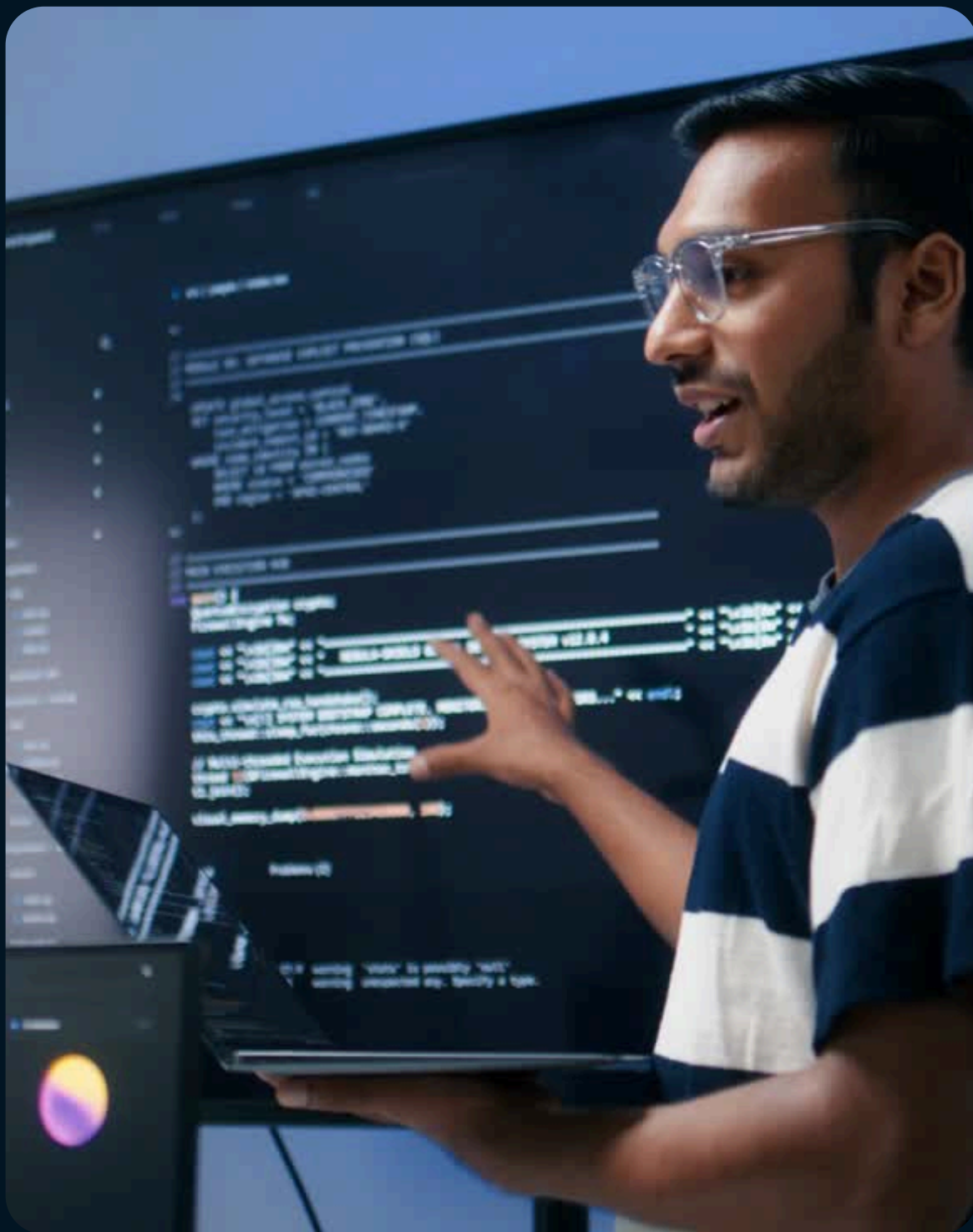
* By Eduardo Rebollar *



Do AI tools affect wages differently depending on developer experience?

== Motivation ==

- Tech and AI are innovating rapidly
- AI tools rapidly changing workflows
- Who adopts AI tech early? Developers!
- Is there merit to learning AI tech?
 - ↑ Productivity / Wages / Profits
- AI's Value ∞ Experience Level
- CS + Econ



◀ Prior Literature ▶

Artificial Intelligence, Automation and Work (2018)

- AI can **substitute** for labor in routine tasks or **complement** workers in complex, non-routine tasks
- Effect on wages depend on the force that is bigger in magnitude
- Theory can't predict what might happen, requires empirical testing

The Impact of AI on Developer Productivity (2023)

- With 95 random developers, Copilot users finished writing an HTTP server in JavaScript 55.8% faster than the control group

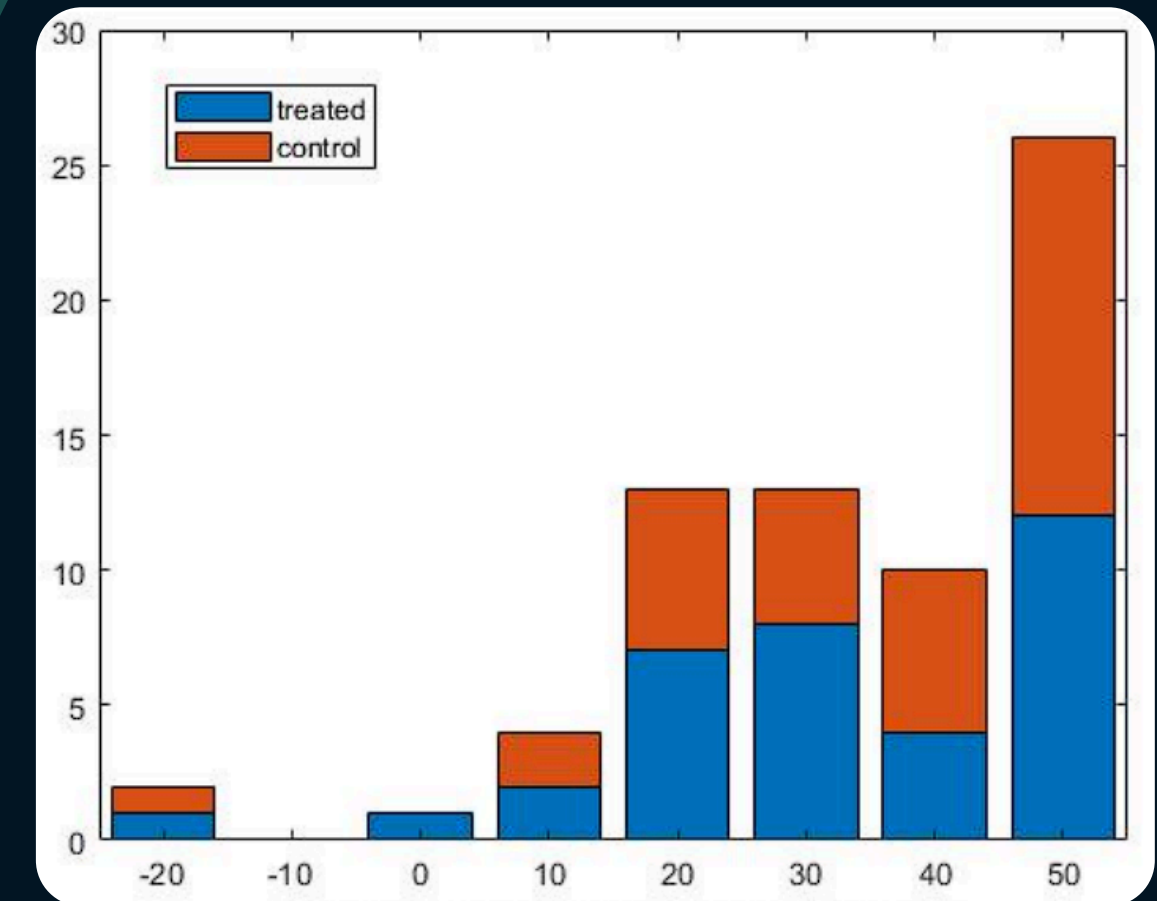
Generative AI at Work (2023)

- With 5000+ customer service agents, 14% average productivity gain, but 34% for novice workers vs. near-zero for experienced workers

The Productivity Effects of Generative AI (2024)

- With 2000+ Microsoft & Accenture developers, 13–22% more pull requests per week at Microsoft; 7–9% at Accenture

The Impact of AI on Developer Productivity



Distribution of Anticipated/Self-Estimated Productivity Gain (%) with Copilot

On average, perceived a 35% ↑
in productivity, in actuality was a
55.8% ↑

2025 Stack Overflow Annual Developer Survey

About this Dataset:

- “Definitive report on the state of software development”
- Received over 49,000+ responses from 177 countries
- Simple to collect, just downloaded the data as a raw CSV file
- Cross-Sectional Data including experience, education, AI tool use, etc, all variables I need in one place
 - No supplemental data sources needed

Decisions:

- Dropped 160+ irrelevant columns; kept variables tied to my model
- Removed observations with missing/zero/not applicable responses
- Removed outliers: 0 and 100 years of work experience
- Removed full-time students, freelancers, retired individuals

Do AI tools affect wages differently depending on developer experience?

$$\ln(\text{Salary}_i) = \beta_0 + \beta_1 \text{AI}_i + \beta_2 \text{Exp}_i + \beta_3 (\text{AI}_i \times \text{Exp}_i) + \beta_4 \text{Edu}_i + \beta_5 \text{Size}_i + \beta_6 \text{Industry}_i + \beta_7 \text{Country}_i + \varepsilon_i$$

Definitions:

- $\ln(\text{Salary}_i)$: log of annual earnings in USD
- AI_i : Dummy = 1 if developer i uses AI tools, 0 otherwise
- Exp_i : Developer i 's years of professional work experience
- $\text{AI}_i \times \text{Exp}_i$: Interaction term
- Controls: Education (Minimum Bachelors), Company Size (Minimum 1000+), Industry (Software Development), Country (Developed/High Income)

Coefficient of Interest (β_3):

- If $\beta_3 > 0$ → AI **complements** Experience (senior developers benefit more)
- If $\beta_3 < 0$ → AI **substitutes** Experience (junior developers benefit more)

Fast Facts

- **Exp_i** ranges from 1 - 58 years (w/ a mean of 13.2)
- 81.4% use AI tools
- 79.3% live in a high-income/developed country
- 81.4% have at least a Bachelor's degree
- 35.2% work for a large company/org
- 45.5% work in the software development industry